Math 231B HW 3

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Problem 1

Etingof Lecture Note Exercise 31.15:

(i) Show that for even $\dim V$, the representations $(\xi^* M)_0, (\xi^* M)_1$ are isomorphic to S_+, S_- respectively.

(ii) Show that for odd $\dim V$, the representations $\xi^* M_+$ and $\xi^* M_-$ are both isomorphic to S.

Hint: Find the highest weight vector for each of these representations and compute the weight of this vector. Then compare dimensions.

Solution:

(i) For even dimension case, recall that V has basis $a_1, ..., a_n, b_1, ..., b_n$, while $M = \bigwedge(a_1, ..., a_n)$. Now, consider the following elements in $\bigwedge^2 V \cong \mathfrak{so}(V)$:

$$\forall i \in \{1, ..., n\}, \quad e_i := a_i \wedge a_{i+1}, \quad f_i := -b_i \wedge b_{i+1}, \quad h_i := [e_i, f_i] \tag{1.1}$$

For the case $i = n$, take $e_n = a_1 \wedge b_n$, $b_n = -b_1 \wedge a_n$ instead. Then, inside the Clifford Algebra $\mathrm{Cl}(V)$ (with Lie algebra homomorphism $\xi : \bigwedge^2 V \to \mathrm{Cl}(V)$ by $\xi(a \wedge b) = \frac{1}{2}(ab - ba) = ab - \frac{1}{2}(a, b))$, their image are:

$$\xi(e_i) = a_i a_{i+1} - \frac{1}{2}(a_i, a_{i+1}) = a_i a_{i+1}, \quad \xi(e_n) = a_n b_1 \tag{1.2}$$

$$\xi(f_i) = -\left(b_i b_{i+1} - \frac{1}{2}(b_i, b_{i+1})\right) = -b_i b_{i+1}, \quad \xi(f_n) = b_n a_1 \tag{1.3}$$

$$\begin{aligned} \xi(h_i) &= \xi([e_i, f_i]) = \xi([a_i \wedge a_{i+1}, -b_i \wedge b_{i+1}]) \\ &= -((a_{i+1}, b_i) a_i b_{i+1} - (a_{i+1}, b_{i+1}) a_i b_i + (a_i, b_i) b_{i+1} a_{i+1} - (a_i, b_{i+1}) b_i a_{i+1}) \\ &= -(-a_i b_i + b_{i+1} a_{i+1}) = a_i b_i + a_{i+1} b_{i+1} - 1 \end{aligned} \tag{1.4}$$

$$\xi(h_n) = \xi([a_1 \wedge b_n, -b_1 \wedge a_n]) = -(b_n a_n + a_1 b_1) = -(a_1 b_1 - a_n b_n + 1) \tag{1.5}$$

Which, the commutation relation of e_i, f_i, h_i in $\mathrm{Cl}(V)$ are given as:

$$
\begin{aligned}
[h_i, e_i] &= [a_i b_i + a_{i+1} b_{i+1}, a_i a_{i+1}] \\
&= a_i a_{i+1} - a_{i+1} a_i \\
&= 2\left(a_i a_{i+1} - \frac{1}{2}(a_i, a_{i+1})\right) \\
&= 2 a_i a_{i+1} = 2 e_i
\end{aligned} \tag{1.6}
$$

$$
\begin{aligned}
[h_i, f_i] &= -[a_i b_i + a_{i+1} b_{i+1}, b_i b_{i+1}] \\
&= -(b_{i+1} b_i - b_i b_{i+1}) \\
&= (b_i b_{i+1} - b_{i+1} b_i) \\
&= 2\left(b_i b_{i+1} - \frac{1}{2}(b_i, b_{i+1})\right) \\
&= 2 b_i b_{i+1} = -2 f_i
\end{aligned} \tag{1.7}
$$

$$
\begin{aligned}
[h_n, e_n] &= [a_n b_n - a_1 b_1, a_1 b_n] \\
&= -b_n a_1 + a_1 b_n \\
&= 2 a_1 b_n = 2 e_n
\end{aligned} \tag{1.8}
$$

$$
\begin{aligned}
[h_n, f_n] &= [a_n b_n - a_1 b_1, -b_1 a_n] \\
&= -(a_n b_1 - b_1 a_n) \\
&= 2 b_1 a_n = -2 f_n
\end{aligned} \tag{1.9}
$$

So, they do satisfy the $\mathfrak{sl}_2$-triple relation.

Now, notice that each $h_i = \left(a_i b_i - \frac{1}{2}\right) + \left(a_{i+1} b_{i+1} - \frac{1}{2}\right)$, and $h_n = \left(a_n b_n - \frac{1}{2}\right) - \left(a_1 b_1 + \frac{1}{2}\right)$. Then, this would generate all $a_i b_i$ and $\frac{1}{2}$ element via addition and scalar multiplication, and further generate all $a_i b_i - \frac{1}{2}$ in our subalgebra. We'll particularly look at the action of each $H_i := a_i b_i - \frac{1}{2}$.

Top Wedge Case:
If consider the action of e_i on the top wedge $(a_1 ... a_n) \in M$, then one has:

$$
\begin{aligned}
&e_i * (a_1 ... a_n) = (a_i a_{i+1}) * (a_1 ... a_n) = 0 \\
&e_n * (a_1 ... a_n) = (a_1 b_n) * (a_1 ... a_n) = a_1 * (-1)^{n-1} (a_1 ... a_{n-1}) = 0
\end{aligned} \tag{1.10}
$$

(Note: Since wedge product with two similar items would automatically be 0).

As a result, if $(a_1 ... a_n)$ belongs to some weight space, it's automatically becomes a highest weight vector.

Now, consider any H_i's action:

$$
\begin{aligned}
H_i * (a_1 ... a_n) &= \left(a_i b_i - \frac{1}{2}\right) * (a_1 ... a_n) \\
&= a_i * (-1)^{i-1} (a_1 ... \hat{a}_i ... a_n) - \frac{1}{2}(a_1 ... a_n) \\
&= (a_1 ... a_n) - \frac{1}{2}(a_1 ... a_n) \\
&= \frac{1}{2}(a_1 ... a_n)
\end{aligned} \tag{1.11}
$$

This shows that with respect to all H_i's, $(a_1...a_n)$ has been acted by multiplying by $\frac{1}{2}$. THis shows that its weight is the fundamental weight $w_{n-1} = \left(\frac{1}{2}, ..., \frac{1}{2}\right)$ for $\mathfrak{so}(2n)$.

Finally, notice hat since any $b_i b_j = \xi(b_i \wedge b_j)$ (for $i \neq j$), then no matter if $a_1...a_n \in (\xi^* M)_0$ or $(\xi^* M)_1$ (corresponds to the case where n is even / odd respectively), it generates everything, as if $a_{i_1}...a_{i_k}$ is a wedge with $n-k$ being even (i.e. they have the same parity), if $a_{j_1}, ..., a_{j_l}$ are distinct variables that didn't show up in $a_{i_1}...a_{i_k}$ (say $l = n-k$), then notice the following:

$$\prod_{i=1}^{\frac{l}{2}=\frac{n-k}{2}} b_{j_{2i}} b_{j_{2i+1}} * (a_1...a_k) = (-1)^L \left(a_{i_1}...a_{i_l}\right) \tag{1.12}$$

since $b_{j_1}...b_{j_l}$ cancels all appearance of $a_{j_1}, ..., a_{j_l}$ in $(a_1...a_n)$, and left with $a_{i_1}...a_{i_k}$ (up to some scalars). So, $a_1...a_n$ is indeed a highest weight vector of $(\xi^* M)_0$ or $(\xi^* M)_1$ with fundamental weight $w_{n-1} = \left(\frac{1}{2}, ..., \frac{1}{2}\right)$ (dependent on the even / oddness of n), and generates everything).

Second Top Wedge Case:

If consider the second top wedge $a_1...a_{n-1}$ (if $a_1...a_n$ belongs to one of $(\xi^* M)_0, (\xi^* M)_1$, then $a_1...a_{n-1}$ must be in the other), notice that each e_i acts as below:

$$\begin{aligned} e_i * (a_1...a_{n-1}) &= (a_i a_{i+1}) * (a_1...a_{n-1}) = 0 \\ e_n * (a_1...a_{n-1}) &= (a_1 b_n) * (a_1...a_{n-1}) = 0 \end{aligned} \tag{1.13}$$

(Note: the first one has $(n+1)$ element in the wedge, which is automatically 0; the second one has b_n acts as $\frac{\partial}{\partial a_n}$, but $a_1...a_{n-1}$ contains none of these).

So, if $(a_1...a_{n-1})$ is in a weight space, it's a highest weight vector.

Now, consider the action of each H_i on this:

$$H_i * (a_1...a_{n-1}) = \left(a_i b_i - \frac{1}{2}\right) * (a_1...a_{n-1}) = \begin{cases} \frac{1}{2}(a_1...a_{n-1}) & i < n \\ -\frac{1}{2}(a_1...a_{n-1}) i = n \end{cases} \tag{1.14}$$

(Note: the reason is because if $i < n$, then a_i shows up in $a_1...a_{n-1}$, so $a_i b_i$ acts as 1; on the other hand, if $i = n$, $a_n b_n$ acts on $a_1...a_{n-1}$ as 0, since it doesn't have a_n term, so the $-\frac{1}{2}$ in H_i takes over).

Hence, $(a_1...a_{n-1})$ in fact corresponds to the weight $w_n = \left(\frac{1}{2}, ..., \frac{1}{2}, -\frac{1}{2}\right)$ for $\mathfrak{so}(2n)$.

Finally, this wedge also generates everything with the same degree parity: if take any other $a_1...\hat{a}_i...a_n$ (where $i \neq n$), then one has $(a_n b_i) * (a_1...a_{n-1}) = (-1)^k (a_1...\hat{a}_i...a_n)$ (by cancelling a_i, and add a_n, up to a scalar).

Also, similar to the top wedge case, if any other wedge is with even degree difference with $a_1...a_{n-1}$, first move to the suitable $a_1...\hat{a}_i...a_n$ (such that the desired wedge also misses a_i), then lower using the combinations of $b_j b_k$'s.

This shows that if $(a_1...a_{n-1})$ lies in $(\xi^* M)_0$ or $(\xi^* M)_1$ (again, based on the odd/evenness of n), it's a highest weight of fundamental weight w_n that generates everything.

Finally, for the above two cases, notice that $\dim(\xi^* M)_0 = \sum_{0 \leq 2k \leq n} \binom{n}{2k}$ (sum of all distinct even degree wedges), and $\dim(\xi^* M)_1 = \sum_{0 \leq 2k+1 \leq n} \binom{n}{2k+1}$ (sum of all choices of distinct odd degree wedges). But, if consider the following binomial expansion:

$$(x+1)^n = \sum_{k=0}^{n} \binom{n}{k} x^n \tag{1.15}$$

Plugin $x = -1$, one yields the following:

$$0 = \sum_{k=1}^{n} \binom{n}{k} (-1)^n = \sum_{0 \le 2k \le n} \binom{n}{2k} - \sum_{0 \le 2k+1 \le n} \binom{n}{2k+1} \tag{1.16}$$

This equation shows that $\dim(\xi^* M)_0 = \dim(\xi^* M)_1$. And, with the two direct sums to $M = \bigwedge(a_1 ... a_n)$ (which has dimension 2^n), each of them must have dimension 2^{n-1}, matching up with the dimension of S_+, S_-.

Since the weight of the highest weight representations match up, together with the dimension matches up, one concludes that $(\xi^* M)_0, (\xi^* M)_1$ are isomorphic to S_+, S_- (depending on the parity of n).

(ii) For odd dimension case, given V with basis $a_1, ..., a_n, b_1, ..., b_n, z$ that satisfies $(a_i, a_j) = (b_i, b_j) = (a_i, z) = (b_i, z) = 0$, $(a_i, b_j) = \delta_{ij}$, and $(z, z) = 2$. We'll define the following elements in $\mathfrak{so}(V) \cong \bigwedge^2 V$:

$$\forall i \in \{1, ..., n\}, \quad e_i := a_i \wedge z, \quad f_i := -b_i \wedge z, \quad h_i := [e_i, f_i] \tag{1.17}$$

If consider their image in the Clifford Algebra $\mathrm{Cl}(V)$ (by the map $\xi : \bigwedge^2 v \to \mathrm{Cl}(V)$ by $\xi(a \wedge b) = \frac{1}{2}(ab - ba) = ab - \frac{1}{2}(a, b)$, which is also a Lie algebra homomorphism), one gets the following:

$$\xi(e_i) = \xi(a_i \wedge z) = \frac{1}{2}(a_i z - z a_i) = \left(a_i z - \frac{1}{2}(a_i, z)\right) = a_i z \tag{1.18}$$

$$\xi(f_i) = \xi(-b_i \wedge z) = -\frac{1}{2}(b_i z - z b_i) = -\left(b_i z - \frac{1}{2}(b_i, z)\right) = -b_i z \tag{1.19}$$

$$\begin{aligned} \xi(h_i) &= \xi([e_i, f_i]) = [a_i z, -b_i z] \\ &= -((z, b_i)\xi(a_i \wedge z) - (z, z)\xi(a_i \wedge b_i) + (a_i, b_i)\xi(z \wedge z) - (a_i, z)\xi(b_i \wedge z)) \\ &= -\left(-2\left(a_i b_i - \frac{1}{2}\right)\right) = 2a_i b_i - 1 \end{aligned} \tag{1.20}$$

Here, we'll use e, f, h to represent the element in $\mathrm{Cl}(V)$. Notice that these do satisfy the $\mathfrak{sl}_2$-triple relation:

$$\begin{aligned} [h_i, e_i] &= [2a_i b_i, a_i z] \\ &= 2((b_i, a_i)\xi(a_i \wedge z) - (b_i, z)\xi(a_i \wedge a_i) + (a_i, a_i)\xi(z \wedge b_i) - (a_i, z)\xi(a_i \wedge b_i)) \\ &= 2\xi(a_i \wedge z) = 2a_i z = 2e_i \end{aligned} \tag{1.21}$$

$$\begin{aligned} [h_i, f_i] &= [2a_i b_i, -b_i z] \\ &= -2((b_i, b_i)\xi(a_i \wedge z) - (b_i, z)\xi(a_i \wedge b_i) + (a_i, b_i)\xi(z \wedge b_i) - (a_i, z)\xi(b_i \wedge b_i)) \\ &= -2(z b_i) = 2b_i z = -2f_i \end{aligned} \tag{1.22}$$

So, these three do satisfy the $\mathfrak{sl}_2$-triple relation.

Now, for each case, consider the action of $\frac{h_i}{2}$ on the top wedge $(a_1 ... a_n) \in M_\pm$ respectively:

$$\frac{h_i}{2} * (a_1 ... a_n) = \left(a_i b_i - \frac{1}{2}\right) * (a_1 ... a_n) = (a_1 ... a_n) - \frac{1}{2}(a_1 ... a_n) = \frac{1}{2}(a_1 ... a_n) \quad (1.23)$$

So, each $\frac{h_i}{2}$ acts as a scalar of $\frac{1}{2}$ on the top wedge $(a_1 ... a_n) \in M_{\pm}$, showing it's a weight vector corresponding to weight $w_n = \left(\frac{1}{2}, ..., \frac{1}{2}\right)$ for $\mathfrak{so}(2n+1)$.

On the other hand, the e_i's have the following action:

$$e_i * (a_1 ... a_n) = (a_i z) * (a_1 ... a_n) = (-1)^n a_i * (a_1 ... a_n) = 0 \quad (1.24)$$

So, this top wedge is annihilated by all e_i's, hence is a highest weight vector.

Finally, notice that each f_i satisfies the following:

$$f_i * (a_1 ... a_n) = (b_i z) * (a_1 ... a_n) = \pm(-1)^n b_i * (a_1 ... a_n) = \pm(-1)^{n+i-1} (a_1 ... \hat{a}_i ... a_n) \quad (1.25)$$

(Note: the $\pm$ depends on if $(a_1 ... a_n) \in M_+$ or M_-, due to the action of z being $\pm(-1)^{\ell(w)} w$ for any pure wedge product $w \in M_{\pm}$).

So, each f_i decreases the degree by canceling the a_i term in the wedge product. Hence, for each wedge product $a_{i_1} ... a_{i_k}$, one has $\prod_{j \neq i_1, ..., i_k} f_j * (a_1 ... a_k) = (-1)^L \left(a_{i_1} ... a_{i_k}\right)$ for some $L \in \mathbb{N}$ (since the operator $\prod_{j \neq i_1, ..., i_k} f_j$ cancels all variable a_j not existing in $i_1, ..., i_k$).

Hence, $(a_1 ... a_n)$ in fact generates the whole $M_{\pm}$. Since each has dimension 2^n, and it's a highest weight module for fundamental weight $w_n = \left(\frac{1}{2}, ..., \frac{1}{2}\right)$, then both $M_{\pm}$ must agree with the spin representation S.

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Problem 2

Etingof Lecture Notes Exercise 35.8 (i), (ii):

Let $G = \mathrm{GL}_n(\mathbb{C})$. A **regular algebraic function** on G is a polynomial of X_{ij} and $\det(X)^{-1}$ for $X \in G$. Denote by $\mathcal{O}(G)$ the algebra of regular algebraic functions on G.

(i) Show that $G \times G$ acts on $\mathcal{O}(G)$ by left and right multiplication.

(ii) (Algebraic Peter-Weyl theorem) Show that as a $G \times G$-module, we have

$$\mathcal{O}(G) = \bigoplus_{V \in \mathrm{Irrep}(G)} V \otimes V^* \tag{2.1}$$

Hint: Compute $\mathrm{Hom}_G(V, \mathcal{O}(G))$ where G acts on $\mathcal{O}(G)$ by right translations. For this, interpret elements of this space as equivariant functions $G \to V^*$ and show that such functions are automatically regular algebraic.

Solution:

(i) Given any pair $(A, B) \in G \times G$, and algebraic function $f(X) \in \mathcal{O}(G)$ (which f is a polynomial with indeterminates X_{ij} and $\det(X)^{-1}$, for any $X \in G = \mathrm{GL}_n(\mathbb{C})$), define the action $((A, B) * f)(X) := f(A^{-1}XB)$.

First, to show it's well-defined, notice that both A^{-1}, B are $n \times n$ matrices, hence for any $X \in \mathrm{GL}_n(\mathbb{C})$, one has the entries $(A^{-1}XB)_{il} = \sum_{j=1}^n \sum_{k=1}^n (A^{-1})_{ij} X_{jk} B_{kl}$, which the entries of $A^{-1}XB$ are all linear combinations of the entris of X; on the other hand, one has $\det(A^{-1}XB)^{-1} = (\det(A) \cdot \det(B)^{-1})\det(X)^{-1}$, which is a multiple of $\det(X)^{-1}$. Hence, with f being a polynomial in entries of the matrix and its determinant's inverse, one has $f(A^{-1}XB)$ still be a polynomial in entries of X and $\det(X)^{-1}$ (since inputs of f now becomes linear combinations of X_{ij}'s, and multiples of $\det(X)^{-1}$, which is still a polynomial in terms of X_{ij}'s and $\det(X)^{-1}$).

Then, to show it's an action, for any $(A, B), (A', B') \in G \times G$, one has the following:

$$\begin{aligned} ((A, B) * ((A', B') * f))(X) &= ((A', B') * f)(A^{-1}XB) = f(A'^{-1}(A^{-1}XB)B') \\ &= f\big((AA')^{-1}X(BB')\big) = ((AA', BB') * f)(X) \\ &= (((A, B) \cdot (A', B')) * f)(X) \end{aligned} \tag{2.2}$$

This shows that $(A, B) * ((A', B') * f) = ((A, B) \cdot (A', B')) * f$, which it's in fact a (left) group action.

(ii) Notice that G is a quasi-affine variety, and V^* is an affine variety, for any V that's a finite-dimensional irreducible representation of V.

First, recall that $\mathrm{Hom}_G(V, \mathcal{O}(G))$ are all linear maps preserving the G-representations. As a result, if look at the symmetric tensor algebra SV, notice that it is equivalent to $\mathcal{O}(V^*)$ (all algebraic functions on V^*): If viewing V^* as some traditional $\mathbb{C}$-vector space, like $\mathbb{C}^m$, choose a basis $x_1, ..., x_m \in V$ and the corresponding dual basis $\varphi_1, ..., \varphi_m \in V^*$, then each linear functional $\varphi \in V^*$ has a unique representative $\varphi = a_1\varphi_1 + ... + a_m\varphi_m$, which $x_i : V^* \to \mathbb{C}$ by $x_i(\varphi) := \varphi(x_i) = a_i\varphi_i(x_i) = a_i$ is the „degree 1 polynomial" associated to it.

As a result, $SV = \mathbb{C}[x_1, ..., x_m]$ is precisely the coordinate ring of $V^* \cong \mathbb{C}^m$ (or $\mathbb{C}[x_1, ..., x_m] \cong \mathcal{O}(V^*)$). Hence, with each morphism in $\mathrm{Hom}_G(V, \mathcal{O}(G))$ indicates a specific way of mapping $x_1, ..., x_m$ into $\mathcal{O}(G)$ (which is in fact a $\mathbb{C}$-algebra, as it's a polynomial algebra over $\mathbb{C}$ on the quasi-affine variety G), hence they each correspond to a unique algebra homomorphism $\mathbb{C}[x_1, ..., x_m] = \mathcal{O}(V^*) \to \mathcal{O}(G)$, that preserves the G-action on the tensor algebra $S = \mathbb{C}[x_1, ..., x_m]$ and on $\mathcal{O}(G)$ (since the morphism mapping $x_1, ..., x_m$

to $\mathcal{O}(G)$ preserves each generateor's G-action). So, there is a one-to-one correspondance $\mathrm{Hom}_G(V, \mathcal{O}(G)) \leftrightarrow \mathrm{Hom}_{\substack{\mathbb{C}-\mathrm{Alg} \\ G-\mathsf{Set}}}(\mathcal{O}(V^*), \mathcal{O}(G))$ (indicating all $\mathbb{C}$-algebra homomorphism between $\mathcal{O}(V^*)$ and $\mathcal{O}(G)$, that is also a G-equivariant map).

Now, notice that this has a one-to-one correspondance to morphisms $\mathrm{Hom}_{G-\mathsf{Set}}(G, V^*)$, where this is also a subset of the morphisms between affine varieties (since all the morphism $f : G \to V^*$ that satisfies $f^* \in \mathrm{Hom}_{\substack{\mathbb{C}-\mathrm{Alg} \\ G-\mathsf{Set}}}(\mathcal{O}(V^*), \mathcal{O}(G))$ must satisfy $f^*(g)(XA) = A * f^*(g)(X) = f^*(g \circ \rho_V(A)^{-1})(X)$ for all $g \in \mathcal{O}(V^*)$, and $A, X \in G$ for the G-action property, so one has $g \circ f(XA) = (g \circ \rho_V(A)^{-1})(f(X)) = (A * g)(f(X))$, in particular showing that f must be a G-equivariant map). So, this establishes a vijection between $\mathrm{Hom}_G(V, \mathcal{O}(G)) \leftrightarrow \mathrm{Hom}_{G-\mathsf{Set}}(G, V^*)$. However, by the irreducibility, one simply has $\mathrm{Hom}_{G-\mathsf{Set}}(G, V^*) \cong V^*$, so one can generate an isomorphism $V \otimes V^* \cong V \otimes \mathrm{Hom}_{G-\mathsf{Set}}(G, V^*) \cong V \otimes \mathrm{Hom}_G(V, \mathcal{O}(G))$.

As a result, there is a natura evaluation $\bigoplus_{V \in \mathrm{Irrep}(G)} V \otimes V^* \cong \bigoplus_{V \in \mathrm{Irrep}(G)} V \otimes \mathrm{Hom}_G(V, \mathcal{O}(G)) \hookrightarrow \mathcal{O}(G)$ via the map $v \otimes f \mapsto f(v)$, for any $v \otimes f \in V \otimes \mathrm{Hom}_G(V, \mathcal{O}(G))$ (where the injection is by the assumption of each component's irreducibility).

Yet, this map is also a surjection, because for any polynomial $f \in \mathcal{O}(G)$, which has a bounded degree in both $\det(X)^{-1}$, and the total degree of X_{ij}'s. Then, if one collects $\mathrm{span}\{g \cdot f \in \mathcal{O}(G) \mid g \in G\}$, notice that every $g \cdot f$ must have the same precise degree in $\det(X)^{-1}$ and the total degree of X_{ij}'s (since g scales $\det(X)^{-1}$ linearly, while mapping linear functions of X_{ij}'s into each of the input of f, which both preserve the degrees). Then, in particular $Z = \mathrm{span}\{g \cdot f \in \mathcal{O}(G) \mid g \in G\}$ is a finite-dimensional subspace of $\mathcal{O}(G)$ due to the bounded degree of all polynomials in there. And, because it collects all G-orbits of f (and their scalings), it's actually a finite-dimensional G-representation over $\mathbb{C}$.

Now, using the complete reducibility of the representation of $G = \mathrm{GL}_n(\mathbb{C})$, then $Z \cong \bigoplus_i V_i$ for finitely many irreducible V_i's that are G-representations. As a result, take the subspace $\bigoplus_i V_i \otimes V_i^*$ in $\bigoplus_{V \in \mathrm{Irrep}(G)} V \otimes V^*$, since its evaluation must contain f (because f is in the direct sum of V_i's), then f must appear in the image of the map $\bigoplus_{V \in \mathrm{Irrep}(G)} V \otimes V^* \cong \bigoplus_{V \in \mathrm{Irrep}(G)} V \otimes \mathrm{Hom}_G(V, \mathcal{O}(G)) \hookrightarrow \mathcal{O}(G)$. This shows the surjection.

As a result, $\mathcal{O}(G) \cong \bigoplus_{V \in \mathrm{Irrep}(G)} V \otimes V^*$.